CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2018 Examination | **Paper Code:** 18024 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to 3 decimal places. [3 Marks]

Solution:

Let $f(x) = x^3 + x^2 - 1$.

- For $x = 0$: $f(0) = 0 + 0 - 1 = -1 < 0$
- For $x = 1$: $f(1) = 1 + 1 - 1 = +1 > 0$

Since $f(0) \cdot f(1) < 0$, a real root lies in the open interval **(0, 1)**.

n	a	b	c = (a+b)/2	f(c)	Sign	Updated Interval
1	0.0000	1.0000	0.5000	-0.6250	Negative (< 0)	[0.5000, 1.0000]
2	0.5000	1.0000	0.7500	-0.0156	Negative (< 0)	[0.7500, 1.0000]
3	0.7500	1.0000	0.8750	+0.4355	Positive (> 0)	[0.7500, 0.8750]
4	0.7500	0.8750	0.8125	+0.1965	Positive (> 0)	[0.7500, 0.8125]
5	0.7500	0.8125	0.7812	+0.0872	Positive (> 0)	[0.7500, 0.7812]
6	0.7500	0.7812	0.7656	+0.0350	Positive (> 0)	[0.7500, 0.7656]
7	0.7500	0.7656	0.7578	+0.0095	Positive (> 0)	[0.7500, 0.7578]
8	0.7500	0.7578	0.7539	-0.0031	Negative (< 0)	[0.7539, 0.7578]
9	0.7539	0.7578	0.7559	+0.0032	Positive (> 0)	[0.7539, 0.7559]
10	0.7539	0.7559	0.7549	+0.00002	Positive (≥ 0)	[0.7539, 0.7549]
11	0.7539	0.7549	0.7544	-0.0015	Negative (< 0)	[0.7544, 0.7549]

At iteration 11, interval width is $|0.7549 - 0.7544| = 0.0005 < 0.001$.

Final Converged Root: x ≈ 0.755 (correct to 3 decimal places)

Q2. Define backward differences and prove the operator identity $E\nabla = \nabla E = \Delta$. [3 Marks]

Solution:

Definition: For equispaced points $x_i = x_0 + i \cdot h$, the **Backward Difference Operator (∇)** is defined as:

$$\nabla y_i = y_i - y_{i-1} \quad \text{and} \quad \nabla^k y_i = \nabla^{k-1} y_i - \nabla^{k-1} y_{i-1} \ (k \geq 2)$$

Backward Difference Table for ordered pairs (x_0, y_0) to (x_4, y_4):

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0	–	–	–	–
x_1	y_1	$\nabla y_1 = y_1 - y_0$	–	–	–
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2$	–	–
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3$	$\nabla^3 y_3$	–
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$

Operator Proofs:

1. $E \bullet \nabla f(x) = E\,[f(x) - f(x-h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{E\nabla = \Delta}$
2. $\nabla \bullet E\, f(x) = \nabla\,[f(x+h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{\nabla E = \Delta}$

Hence Proved: $E\nabla = \nabla E = \Delta$

Q3. Find the value of $\frac{dy}{dx}$ at $x = 1.2$ for points (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552)... [3 Marks]

Solution:

Given step size $h = 1.2 - 1.0 = 0.2$. Let $x_0 = 1.2$, $y_0 = 3.3201$.

Neighbors: $x_{-1} = 1.0$, $y_{-1} = 2.7183$ and $x_1 = 1.4$, $y_1 = 4.0552$.

Central Difference Formula (First Derivative):

$$f'(x_0) \approx \frac{y_1 - y_{-1}}{2h}$$

$$f'(1.2) = \frac{4.0552 - 2.7183}{2(0.2)} = \frac{1.3369}{0.40} = 3.34225$$

(Newton's Forward Difference check: $f'(1.2) \approx (1/0.2)[0.7351 - 0.5(0.1627) + (1/3)(0.0361)] = 3.3289$. True derivative $e^{1.2} \approx 3.3201$.)

dy/dx at x = 1.2 is 3.34225

Q4. Solve the following linear system using Gauss Elimination method: [3 Marks]

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$x + y + 3z = 5$$

Solution:

Step 1: Form the Augmented Matrix [A | B]:

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 2 & 10 & 1 & 13 \\ 1 & 1 & 3 & 5 \end{array}\right)$$

Step 2: Forward Elimination to Upper Triangular Form:

- $R_2 \to R_2 - (2/10)R_1 = R_2 - 0.2R_1 \Rightarrow$ [0, 9.8, 0.8 | 10.6]
- $R_3 \to R_3 - (1/10)R_1 = R_3 - 0.1R_1 \Rightarrow$ [0, 0.9, 2.9 | 3.8]
- $R_3 \to R_3 - (0.9 / 9.8)R_2 \Rightarrow$ [0, 0, 2.82653 | 2.82653]

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 0 & 9.8 & 0.8 & 10.6 \\ 0 & 0 & 2.82653 & 2.82653 \end{array}\right)$$

Step 3: Back Substitution:

- From Row 3: 2.82653 • z = 2.82653 ⇒ **z = 1**
- From Row 2: 9.8 • y + 0.8(1) = 10.6 ⇒ 9.8 • y = 9.8 ⇒ **y = 1**
- From Row 1: 10 • x + 1(1) + 1(1) = 12 ⇒ 10 • x = 10 ⇒ **x = 1**

Unique Solution: x = 1, y = 1, z = 1

Q5. Solve $\frac{dy}{dx} = x + y^2$, with y(0) = 1 using Picard's method up to second approximation. [3 Marks]

Solution:

Picard's Successive Approximation Formula:

$$y^{(n)}(x) = y_0 + \int_{x_0}^{x} f(t, y^{(n-1)}(t))\, dt$$

Given initial conditions: $x_0 = 0$, $y_0 = 1$, and $f(t, y) = t + y^2$.

First Approximation ($y^{(1)}$):

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2)\, dt = 1 + \left[\frac{t^2}{2} + t\right]_0^x = 1 + x + \frac{x^2}{2}$$

Second Approximation ($y^{(2)}$):

$$y^{(2)}(x) = 1 + \int_0^x \left[t + (1 + t + \frac{t^2}{2})^2\right] dt$$

$$\text{Expand:} \quad (1 + t + \frac{t^2}{2})^2 = 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4}$$

$$\text{Integrand} = t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} = 1 + 3t + 2t^2 + t^3 + \frac{t^4}{4}$$

$$y^{(2)}(x) = 1 + \left[t + \frac{3t^2}{2} + \frac{2t^3}{3} + \frac{t^4}{4} + \frac{t^5}{20}\right]_0^x$$

$$y^{(2)}(x) = 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5$$

$y^{(1)} = 1 + x + x^2/2$ | $y^{(2)} = 1 + x + (3/2)x^2 + (2/3)x^3 + (1/4)x^4 + (1/20)x^5$

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Find a real root of the equation $2x = \log_{10}(x) + 7$ between 3 and 4 using Method of False Position (Regula Falsi). [7.5 Marks]

Solution:

Let $f(x) = 2x - \log_{10}(x) - 7 = 0$.

- For $x_0 = 3$: $f(3) = 2(3) - \log_{10}(3) - 7 = 6 - 0.47712 - 7 = -1.47712 < 0$
- For $x_1 = 4$: $f(4) = 2(4) - \log_{10}(4) - 7 = 8 - 0.60206 - 7 = +0.39794 > 0$

Since $f(3)$ and $f(4)$ have opposite signs, the root lies in the interval **(3, 4)**.

Regula Falsi Iteration Formula:

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$

Iteration 1:

$$x_2 = \frac{3(0.39794) - 4(-1.47712)}{0.39794 - (-1.47712)} = \frac{1.19382 + 5.90848}{1.87506} = \frac{7.10230}{1.87506} \approx 3.78777$$

Evaluate $f(x_2)$:

$f(3.78777) = 2(3.78777) - \log_{10}(3.78777) - 7 = 7.57554 - 0.57838 - 7 = -0.00284 < 0$.

Root lies in **[3.78777, 4.00000]**.

Iteration 2:

$$x_3 = \frac{3.78777(0.39794) - 4(-0.00284)}{0.39794 - (-0.00284)} = \frac{1.50731 + 0.01136}{0.40078} = \frac{1.51867}{0.40078} \approx 3.7893$$

Evaluate $f(x_3)$: $f(3.7893) = 2(3.7893) - \log_{10}(3.7893) - 7 = 7.57860 - 0.57856 - 7 = +0.00004 \approx 0$.

Final Converged Root: x ≈ 3.7893

Q7. Find the missing terms in the following table using finite differences: [7.5 Marks]

x	0	5	10	15	20	25	30
y	1	3	? (y_2)	73	225	? (y_5)	1153

Solution:

Let the missing values be y_2 (at $x = 10$) and y_5 (at $x = 25$).

Since 5 values are given (y_0, y_1, y_3, y_4, y_6), the data can be represented by a polynomial of degree 4.

Therefore, the 5th finite difference is identically zero: $\Delta^5 y_0 = 0$ and $\Delta^5 y_1 = 0$.

Using shift operator $\Delta = E - 1$:

$$(E-1)^5 y = (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1)\, y = 0$$

Equation 1 (applying on y_0):

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0$$
$$y_5 - 5(225) + 10(73) - 10y_2 + 5(3) - 1 = 0$$
$$y_5 - 1125 + 730 - 10y_2 + 15 - 1 = 0$$
$$-10y_2 + y_5 = 381 \qquad \text{— (Equation 1)}$$

Equation 2 (applying on y_1):

$$y_6 - 5y_5 + 10y_4 - 10y_3 + 5y_2 - y_1 = 0$$
$$1153 - 5y_5 + 10(225) - 10(73) + 5y_2 - 3 = 0$$
$$1153 - 5y_5 + 2250 - 730 + 5y_2 - 3 = 0$$
$$5y_2 - 5y_5 + 2670 = 0 \Rightarrow y_2 - y_5 = -534 \Rightarrow y_5 - y_2 = 534 \qquad \text{— (Equation 2)}$$

Solving the Linear System:

From Equation 2: $y_5 = y_2 + 534$.

Substitute into Equation 1: $-10y_2 + (y_2 + 534) = 381 \Rightarrow -9y_2 = 381 - 534 = -153 \Rightarrow \mathbf{y_2 = 17}$.

Then $y_5 = 17 + 534 = \mathbf{551}$.

Missing Values: $y_2 = 17$ (at x = 10) | $y_5 = 551$ (at x = 25)

Q8. Evaluate $I = \int_3^7 x^2 \log_{10}(x)\, dx$ using Simpson's 1/3 rule with $n = 4$. [7.5 Marks]

Solution:

Given limits $a = 3$, $b = 7$, and number of sub-intervals $n = 4$.

Step size: $h = (b - a) / n = (7 - 3) / 4 = \mathbf{1}$.

i	x_i	Function $f(x) = x^2 \log_{10}(x)$	Ordinate y_i
0	3	$3^2 \cdot \log_{10}(3) = 9 \times 0.477121$	4.2941
1	4	$4^2 \cdot \log_{10}(4) = 16 \times 0.602060$	9.6330
2	5	$5^2 \cdot \log_{10}(5) = 25 \times 0.698970$	17.4743
3	6	$6^2 \cdot \log_{10}(6) = 36 \times 0.778151$	28.0134
4	7	$7^2 \cdot \log_{10}(7) = 49 \times 0.845098$	41.4099

Simpson's 1/3 Rule Formula:

$$I = \frac{h}{3}\left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)\right]$$

Sum of Extremes: $y_0 + y_4 = 4.2941 + 41.4099 = \mathbf{45.7040}$

Sum of Odd Ordinates: $4(y_1 + y_3) = 4(9.6330 + 28.0134) = 4(37.6464) = \mathbf{150.5856}$

Sum of Even Ordinates: $2(y_2) = 2(17.4743) = \mathbf{34.9486}$

Total Bracket Sum $= 45.7040 + 150.5856 + 34.9486 = \mathbf{231.2382}$

$$I = \frac{1}{3} \times 231.2382 = \mathbf{77.0794}$$

Evaluated Integral: $I \approx 77.0794$

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Use Newton-Raphson method to find a real root of $\sin(x) = x / 2$ correct to 3 decimal places between π/2 and π. [15 Marks]

Solution:

Let $f(x) = \sin(x) - \frac{x}{2} = 0$.

Differentiating with respect to x: $f'(x) = \cos(x) - 0.5$.

Given interval: [π/2, π] ≈ [1.5708, 3.1416] radians.

Let initial approximation be $\mathbf{x_0 = 2.0}$ **radians**.

Newton-Raphson Iteration Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iteration 1 (at $x_0 = 2.0$):

$$f(2.0) = \sin(2.0) - 1.0 = 0.90930 - 1.0 = \mathbf{-0.09070}$$
$$f'(2.0) = \cos(2.0) - 0.5 = -0.41615 - 0.5 = \mathbf{-0.91615}$$
$$x_1 = 2.0 - \frac{-0.09070}{-0.91615} = 2.0 - 0.09900 = \mathbf{1.90100}$$

Iteration 2 (at x_1 = 1.90100):

$$f(1.90100) = \sin(1.90100) - \frac{1.90100}{2} = 0.94598 - 0.95050 = \mathbf{-0.00452}$$
$$f'(1.90100) = \cos(1.90100) - 0.5 = -0.32422 - 0.5 = \mathbf{-0.82422}$$
$$x_2 = 1.90100 - \frac{-0.00452}{-0.82422} = 1.90100 - 0.00548 = \mathbf{1.89552}$$

Iteration 3 (at x_2 = 1.89552):

$$f(1.89552) = \sin(1.89552) - \frac{1.89552}{2} = 0.94775 - 0.94776 = \mathbf{-0.00001}$$
$$f'(1.89552) = \cos(1.89552) - 0.5 = -0.31899 - 0.5 = \mathbf{-0.81899}$$
$$x_3 = 1.89552 - \frac{-0.00001}{-0.81899} = 1.89552 - 0.00001 = \mathbf{1.89549}$$

Root: x ≈ 1.895 radians (or ≈ 108.57°)

Q10. Find the Lagrange interpolating polynomial of degree 2 for *y = $\log_e(x)$* from the given table. Hence determine *$\log_e(2.7)$*. [15 Marks]

x	2.0	2.5	3.0
y = $\log_e$(x)	0.69315	0.91629	1.09861

Solution:

Given $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Target value: $x = 2.7$.

Lagrange Interpolation Formula for n = 2:

$$P_2(x) = L_0(x)\, y_0 + L_1(x)\, y_1 + L_2(x)\, y_2$$

Calculating Lagrange Cardinal Polynomials at x = 2.7:

$$L_0(2.7) = \frac{(2.7-2.5)(2.7-3.0)}{(2.0-2.5)(2.0-3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.50} = \mathbf{-0.12}$$
$$L_1(2.7) = \frac{(2.7-2.0)(2.7-3.0)}{(2.5-2.0)(2.5-3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = \mathbf{+0.84}$$
$$L_2(2.7) = \frac{(2.7-2.0)(2.7-2.5)}{(3.0-2.0)(3.0-2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.50} = \mathbf{+0.28}$$

Verification: $L_0 + L_1 + L_2 = -0.12 + 0.84 + 0.28 = 1.00$ ✓

Evaluating $P_2(2.7)$:

$$P_2(2.7) = (-0.12)(0.69315) + (0.84)(0.91629) + (0.28)(1.09861)$$
$$= -0.083178 + 0.769684 + 0.307611$$
$$= \mathbf{0.994117 \approx 0.99412}$$

(Exact analytical value: ln(2.7) ≈ 0.99325; error ≈ 0.00087.)

Interpolated Value: $\log_e(2.7) \approx 0.99412$

Q11. Use Simpson's 3/8 rule to evaluate $\int_0^1 \frac{dx}{1+x}$ with $h = 1/6$. [15 Marks]

Solution:

Given $a = 0$, $b = 1$, step size $h = 1/6 \Rightarrow n = (1 - 0)/(1/6) = 6$ subintervals.

Integrand: $f(x) = \frac{1}{1+x}$.

i	x_i	Exact $y_i = 1 / (1 + x_i)$	Decimal Value
0	0	1 / 1	1.000000
1	1/6	1 / (7/6) = 6/7	0.857143
2	2/6	1 / (8/6) = 6/8 = 3/4	0.750000
3	3/6	1 / (9/6) = 6/9 = 2/3	0.666667
4	4/6	1 / (10/6) = 6/10 = 3/5	0.600000
5	5/6	1 / (11/6) = 6/11	0.545455
6	1	1 / 2	0.500000

Simpson's 3/8 Rule Formula:

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)\right]$$

Extremes: $y_0 + y_6 = 1.000000 + 0.500000 = \mathbf{1.500000}$

Non-multiples of 3: $3(y_1 + y_2 + y_4 + y_5) = 3(0.857143 + 0.750000 + 0.600000 + 0.545455) = 3(2.752598)$
$= \mathbf{8.257794}$

Multiples of 3: $2(y_3) = 2(0.666667) = \mathbf{1.333334}$

Sum of brackets $= 1.500000 + 8.257794 + 1.333334 = \mathbf{11.091128}$

$$I = \frac{3(1/6)}{8} \times 11.091128 = \frac{1/2}{8} \times 11.091128 = \frac{11.091128}{16} = \mathbf{0.693196}$$

(Exact analytical integral: $\int_0^1 dx/(1+x) = [\ln(1+x)]_0^1 = \ln(2) \approx 0.693147$. Highly accurate to 4 decimal places.)

Evaluated Integral: I ≈ 0.693196

Q12. Solve the following linear system using Gauss-Seidel iterative method: [15 Marks]

$$6x + y + z = 20$$
$$x + 4y - z = 6$$
$$x - y + 5z = 7$$

Solution:

Step 1: Check Diagonal Dominance Condition:

- Equation 1: $|6| > |1| + |1| = 2$ (Dominant in x ✓)
- Equation 2: $|4| > |1| + |-1| = 2$ (Dominant in y ✓)
- Equation 3: $|5| > |1| + |-1| = 2$ (Dominant in z ✓)

The system is strictly diagonally dominant. Convergence is guaranteed.

Step 2: Express in Iterative Form:

$$x^{(k+1)} = \frac{20 - y^{(k)} - z^{(k)}}{6}$$
$$y^{(k+1)} = \frac{6 - x^{(k+1)} + z^{(k)}}{4}$$
$$z^{(k+1)} = \frac{7 - x^{(k+1)} + y^{(k+1)}}{5}$$

Step 3: Iteration Table (Initial Approximation: $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$):

Iteration (k)	x = (20 – y – z) / 6	y = (6 – x + z) / 4	z = (7 – x + y) / 5
1	(20 - 0 - 0)/6 = **3.3333**	(6 - 3.3333 + 0)/4 = **0.6667**	(7 - 3.3333 + 0.6667)/5 = **0.8667**
2	(20 - 0.6667 - 0.8667)/6 = **3.0778**	(6 - 3.0778 + 0.8667)/4 = **0.9472**	(7 - 3.0778 + 0.9472)/5 = **0.9739**
3	(20 - 0.9472 - 0.9739)/6 = **3.0131**	(6 - 3.0131 + 0.9739)/4 = **0.9902**	(7 - 3.0131 + 0.9902)/5 = **0.9954**
4	(20 - 0.9902 - 0.9954)/6 = **3.0024**	(6 - 3.0024 + 0.9954)/4 = **0.9983**	(7 - 3.0024 + 0.9983)/5 = **0.9992**
5	(20 - 0.9983 - 0.9992)/6 = **3.0004**	(6 - 3.0004 + 0.9992)/4 = **0.9997**	(7 - 3.0004 + 0.9997)/5 = **0.9999**
6	**3.0000**	**1.0000**	**1.0000**

Converged Solution: x = 3, y = 1, z = 1

Q13. Using Fourth-Order Runge-Kutta method, find *y(0.1)* and *y(0.2)* correct to 4 decimal places, given: [15 Marks]

$$\frac{dy}{dx} = y - x, \quad \text{with initial condition } y(0) = 2$$

Solution:

Given $f(x, y) = y - x$, initial point $x_0 = 0$, $y_0 = 2$, step size $h = 0.1$.

Part 1: Calculation of y(0.1) from $x_0 = 0$, $y_0 = 2$:

$$k_1 = h \bullet f(x_0, y_0) = 0.1(2 - 0) = \mathbf{0.200000}$$
$$k_2 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 0.1 \bullet f(0.05, 2.10) = 0.1(2.10 - 0.05) = \mathbf{0.205000}$$
$$k_3 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) = 0.1 \bullet f(0.05, 2.1025) = 0.1(2.1025 - 0.05) = \mathbf{0.205250}$$
$$k_4 = h \bullet f(x_0 + h, y_0 + k_3) = 0.1 \bullet f(0.1, 2.20525) = 0.1(2.20525 - 0.1) = \mathbf{0.210525}$$
$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$= \frac{1}{6}(0.200000 + 0.410000 + 0.410500 + 0.210525) = \frac{1.231025}{6} \approx \mathbf{0.205171}$$
$$y(0.1) = y_0 + k = 2 + 0.205171 = 2.2052$$

Part 2: Calculation of y(0.2) from $x_1 = 0.1$, $y_1 = 2.205171$:

$k_1 = h \bullet f(x_1, y_1) = 0.1(2.205171 - 0.1) = \mathbf{0.210517}$

$k_2 = h \bullet f(0.1 + 0.05, 2.205171 + 0.105258) = 0.1 \bullet f(0.15, 2.310430) = 0.1(2.310430 - 0.15) = \mathbf{0.216043}$

$k_3 = h \bullet f(0.1 + 0.05, 2.205171 + 0.108022) = 0.1 \bullet f(0.15, 2.313193) = 0.1(2.313193 - 0.15) = \mathbf{0.216319}$

$k_4 = h \bullet f(0.1 + 0.1, 2.205171 + 0.216319) = 0.1 \bullet f(0.2, 2.421490) = 0.1(2.421490 - 0.2) = \mathbf{0.222149}$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.210517 + 0.432086 + 0.432638 + 0.222149) = \frac{1.297390}{6} \approx \mathbf{0.216232}$$

$$y(0.2) = y_1 + k = 2.205171 + 0.216232 = 2.4214$$

y(0.1) = 2.2052 | y(0.2) = 2.4214